JUNIOR SCHOOL LEARNING HUB

GRADE 9 AGRICULTURE
END-OF-TERM 2 EXAMINATION

Name: _____	Adm. No.: _____	Date: _____	Class: _____
Time: 2 hours 30 minutes	Total: 100 marks	Term: 2	Learning Area: Agriculture

INSTRUCTIONS TO LEARNERS

- Answer all questions in Sections A and B.
- Write clearly and use correct agricultural terms.
- Show calculations where required.
- Observe safety when describing or carrying out agricultural activities.

SECTION A: SHORT-ANSWER QUESTIONS (40 MARKS)

1. Define soil fertility. (2 marks)

2. State two ways of maintaining soil fertility on a farm. (2 marks)

3. Name two signs of soil erosion. (2 marks)

4. State two ways of controlling soil erosion on sloping land. (2 marks)

5. Give two examples of organic manure. (2 marks)

6. State two benefits of crop rotation. (2 marks)

7. Name two factors considered when selecting seeds for planting. (2 marks)

8. State two conditions necessary for seed germination. (2 marks)

9. Give two reasons for thinning seedlings. (2 marks)

10. State two advantages of mulching. (2 marks)

11. Name two common farm tools used for land preparation. (2 marks)

12. State two safety precautions when using farm tools. (2 marks)

13. Define irrigation. (2 marks)

14. Name two methods of irrigation. (2 marks)

15. State two signs that a crop is ready for harvesting. (2 marks)

16. Give two ways of reducing post-harvest losses. (2 marks)

17. Name two livestock products obtained from cattle. (2 marks)

18. State two characteristics of a healthy farm animal. (2 marks)

19. Give two reasons for keeping farm records. (2 marks)

20. State two ways of conserving water on a farm. (2 marks)

SECTION B: STRUCTURED QUESTIONS (60 MARKS)

21. Soil management and conservation (10 marks)

a) Explain the meaning of soil erosion. (2 marks)

b) Describe three practices that can reduce soil erosion on a sloping farm. (3 marks)

c) State three benefits of adding organic matter to soil. (3 marks)

d) Give two reasons why soil testing is useful to a farmer. (2 marks)

22. Crop establishment and field management (10 marks)

a) Distinguish between direct sowing and transplanting. (2 marks)

b) State three qualities of good planting materials. (3 marks)

c) Explain two reasons for weeding crops. (2 marks)

d) State three factors that determine the spacing of crops. (3 marks)

23. Water management and irrigation (10 marks)

a) Explain why water is important to crop growth. Give three reasons. (3 marks)

b) Describe two methods of irrigation and how each supplies water. (4 marks)

c) State three practices that help conserve water on a farm. (3 marks)

24. Livestock management and health (10 marks)

a) State four signs of ill health in farm animals. (4 marks)

b) Explain two measures used to prevent livestock diseases. (4 marks)

c) State two reasons for keeping livestock housing clean and dry. (2 marks)

25. Harvesting, handling and storage (10 marks)

a) State three factors that determine the right time to harvest a crop. (3 marks)

b) Explain three causes of post-harvest losses. (3 marks)

c) Describe two appropriate ways of storing harvested grains. (4 marks)

26. Farm records and enterprise planning (10 marks)

a) Name four types of farm records. (4 marks)

b) State three ways farm records help a farmer. (3 marks)

c) A farmer harvests 240 kg of beans and sells them at KSh 80 per kilogram. Calculate the total sales revenue. (2 marks)

d) State one reason for preparing a simple farm budget before starting an enterprise. (1 mark)

MARKING SCHEME

Grade 9 Agriculture – End-of-Term 2 Examination

Total: 100 marks | Section A: 40 marks | Section B: 60 marks

SECTION A (40 MARKS)

1. The ability of soil to supply essential nutrients in suitable amounts and forms for healthy plant growth. (2)

2. Any two: applying manure/compost; crop rotation with legumes; cover cropping; appropriate fertiliser use; agroforestry. (2)

3. Any two: rills/gullies; exposed roots; loss of topsoil; deposition of soil downhill; formation of channels. (2)

4. Any two: terracing; contour farming; grass strips; cover crops; mulching; tree planting. (2)

5. Any two: farmyard manure, compost, green manure, poultry manure. (2)

6. Any two: breaks pest/disease cycles; maintains fertility; improves soil structure; supports better yields; manages weeds. (2)

7. Any two: viability/germination rate; purity; freedom from pests/diseases; variety suitability; seed size/condition. (2)

8. Any two: water, oxygen and suitable temperature. (2)

9. Any two: reduce competition; provide adequate space; improve light/air circulation; promote healthy growth. (2)

10. Any two: conserves moisture; suppresses weeds; reduces erosion; moderates soil temperature; adds organic matter if biodegradable. (2)

11. Any two: hoe, jembe, fork, spade, rake, plough. (2)

12. Any two: use the correct tool; inspect before use; wear protective clothing; carry safely; store properly; keep safe distance. (2)

13. Artificial application/supply of water to crops when rainfall is inadequate. (2)

14. Any two: drip, sprinkler, furrow, basin, flood irrigation. (2)

15. Any two: change in colour; full size/maturity; dry pods/leaves; firm grains; crop-specific maturity signs. (2)

16. Any two: careful harvesting/handling; timely drying; proper storage; pest control; sorting and packaging. (2)

17. Any two: milk, meat, hides/skins, manure. (2)

18. Any two: alertness; good appetite; smooth coat; clear eyes; normal movement; normal droppings; no abnormal discharge. (2)

19. Any two: monitor performance; plan and make decisions; calculate profit/loss; track inputs and outputs; provide evidence of transactions. (2)

20. Any two: rainwater harvesting; mulching; drip irrigation; repair leaks; use water-efficient methods; maintain water pans. (2)

SECTION B (60 MARKS)

21. Soil management and conservation (10 marks)

The detachment and movement/removal of topsoil by agents such as water or wind. (2)

Any three explained: terracing, contour ploughing, strip cropping, grass strips, cover crops, planting trees, mulching. (1 each)

Improves soil structure; increases water-holding capacity; supplies nutrients; supports soil organisms; reduces erosion. Any three. (1 each)

Identifies nutrient/pH status; guides appropriate fertiliser/manure application; avoids waste and improves yields. Any two. (1 each)

22. Crop establishment and field management (10 marks)

Direct sowing: seeds planted in the field where crop grows; transplanting: seedlings raised elsewhere then moved to field. (2)

Healthy, disease-free, viable, true-to-type, mature, undamaged and suitable variety. Any three. (1 each)

Reduces competition for water, nutrients, light and space; removes weed hosts; eases field operations. Any two explained. (1 each)

Crop variety/plant size; soil fertility; rainfall/water availability; intended use; machinery/intercropping system. Any three. (1 each)

23. Water management and irrigation (10 marks)

Photosynthesis; transport of nutrients; seed germination; maintaining turgidity; cooling/transpiration. Any three. (1 each)

Any two: drip (water delivered near roots through emitters); sprinkler (water sprayed like rainfall); furrow (water flows in channels); basin/flood (water covers a level area). 2 marks each. (4)

Mulching; rainwater harvesting; drip irrigation; fixing leaks; planting drought-tolerant crops; timely irrigation; conserving wetlands. Any three. (1 each)

24. Livestock management and health (10 marks)

Loss of appetite; dullness; abnormal discharges; diarrhoea; coughing; lameness; rough coat; unusual isolation; abnormal temperature. Any four. (4)

Vaccination; hygiene/clean housing; quarantine new or sick animals; parasite control; proper nutrition; veterinary checks. Explain any two. (2 each)

Reduces disease and parasites; improves comfort/productivity; reduces injuries and bad odours. Any two. (2)

25. Harvesting, handling and storage (10 marks)

Maturity indicators; intended market/use; weather; market demand; crop variety; moisture content. Any three. (1 each)

Poor handling/bruising; pests and diseases; excess moisture; unsuitable storage; delayed harvesting; contamination. Any three explained. (1 each)

Dry grain to safe moisture; store in clean, dry, ventilated, pest-proof containers/granary; raise bags off floor; inspect regularly. Any two described. (2 each)

26. Farm records and enterprise planning (10 marks)

Production, sales, purchase, labour, inventory/stock, input-use, breeding, health, financial/cashbook records. Any four. (4)

Track income and expenditure; monitor production; plan purchases; identify profitable enterprises; support decision-making; provide evidence for credit/tax where applicable. Any three. (1 each)

240 × 80 = KSh 19,200. (2)

Estimates costs and expected income; helps plan resources and assess viability. (1)

ADMINISTRATION NOTE

This is an original teacher-prepared assessment resource for practice and school-based assessment. Before administration, the Agriculture teacher should confirm that the assessed sub-strands and practical skills have been covered in the actual Grade 9 Term 2 teaching sequence and the school scheme of work. Accept other scientifically and agriculturally correct responses where appropriate.

Prepared by: Junior School Learning Hub